DEPARTEMENT STATISTIEK EN AKTUARIËLE WETENSKAP
DEPARTEMENT OF STATISTICS AND ACTUARIAL SCIENCE

STATISTIEK EN DATAWETENSKAP 188
STATISTICS AND DATA SCIENCE 188

TUTORIAAL 10 / *TUTORIAL* 10

Probleem 1 / *Problem 1*

(a) Saamgevoegde-variansie / *Pooled variance*:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(8 - 1) \times 4^2 + (15 - 1) \times 5^2}{8 + 15 - 2} = 22$$

t-toetsstatistiek / *t-test statistic*:

$$t_{stat} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(42 - 34) - 0}{\sqrt{22 \times \left(\frac{1}{8} + \frac{1}{15}\right)}} \approx 3.8959$$

(b) Vryheidsgrade / *Degrees of freedom*:

$$\text{df} = n_1 + n_2 - 2 = 8 + 15 - 2 = 21$$

(c) Kritieke waarde / *Critical value*:

$$t_{crit} = t_{\text{df}}^{(\alpha)} = t_{21}^{(0.01)} = 2.518$$

(d) Beslissing: Aangesien $t_{stat} = 3.8959 > 2.518 = t_{krit}$, verwerp H_0. Daar is genoeg bewyse om tot die gevolgtrekking te kom dat die eerste populasiegemiddelde groter is as die tweede populasiegemiddelde.
/ *Decision: Since* $t_{stat} = 3.8959 > 2.518 = t_{crit}$, *reject* H_0. *There is enough evidence to conclude that the population mean from the first population is larger than the second population mean.*

(e)
Beslissing: Aangesien *p*-waarde = 0.0004 < 0.01, verwerp H_0. Daar is genoeg bewyse om tot die gevolgtrekking te kom dat die eerste populasiegemiddelde groter is as die tweede populasiegemiddelde.
/ *Decision: Since p*-value = 0.0004 < 0.01, *reject* H_0. *There is enough evidence to conclude that the population mean from the first population is larger than the second population mean.*

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.
Kopiereg voorbehou / *Copyright reserved*

	A	B	C	D
1	**Population 1**		**Population 2**	
2	Sample Size (n_1)	8	Sample Size (n_2)	15
3	Sample Mean ($\bar{x}_1$)	42.00	Sample Mean ($\bar{x}_2$)	34.00
4	Sample Standard Deviation (s_1)	4.00	Sample Standard Deviation (s_2)	5.00
5				
6	**Test Details**		**Excel Commands**	
7	Null Hypothesis (H_0) $\mu_1 - \mu_2 =$	0.00		
8	Alt. Hypothesis (H_1) $\mu_1 - \mu_2 >$	0.00		
9	Significance Level (α)	0.01		
10				
11	**Intermediate Calculations**			
12	Degrees of Freedom (df)	21	= B2 + D2 - 2	
13	Pooled Variance (s_p^2)	22.0000	= (((B2-1) * (B4^2)) + ((D2-1) * (D4^2))) / B12	
14	t test Statistic (t_{stat})	3.8959	= (B3 - D3) / SQRT(B13 * ((1/B2) + (1/D2)))	
15				
16	**One-Tail Hypothesis Test**			
17	p-value	0.0004	= 1 - T.DIST(B14, B12, TRUE)	
18				
19	**Decision**			
20	p-value approach	Reject H0	= IF(B17<B9, "Reject H0", "Fail to reject H0")	

Probleem 2 / *Problem 2*

(a)

Nulhipotese / *Null Hypothesis* (H_0): $\mu_1 - \mu_2 = 0$

Alternatiewe Hipotese / *Alternative Hypothesis* (H_1): $\mu_1 - \mu_2 \neq 0$

Betekenisvlak / *Significance Level* (α): 0.05

Hipotese Toets / *Hypothesis Test*: Tweestert / *Two-tail*

Kritieke waardes / *Critical Values*: $z_{crit} = \pm z_{0.025} = \pm 1.96$

Toetsstatistiek / *Test Statistic* (z_{stat}):

$$z_{stat} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{(375 - 362) - 0}{\sqrt{\frac{110^2}{25} + \frac{125^2}{25}}} \approx 0.3904$$

Beslissing: Aangesien $z_{stat} \approx 0.3904$ tussen die kritieke waardes ± 1.96 is, kan H_0 nie verwerp word nie. Daar is nie genoeg bewyse om tot die gevolgtrekking te kom dat die gloeilampe wat deur Masjien I vervaardig is 'n verskillende populasiegemiddelde leeftyd het as dié gloeilampe wat deur Masjien II vervaardig is nie.

/ *Decision: Since* $z_{stat} \approx 0.3904$ *is between the critical bounds of* ± 1.96*, do not reject* H_0*. There is not enough evidence to conclude that light bulbs produced by Machine I have different population average life times from light bulbs produced by Machine II.*

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.
Kopiereg voorbehou / *Copyright reserved*

(b)

	A	B	C	D
1	**Population 1**		**Population 2**	
2	Sample Size (n_1)	25	Sample Size (n_2)	25
3	Sample Mean ($\bar{x}_1$)	375.00	Sample Mean ($\bar{x}_2$)	362.00
4	Population Std. Deviation (σ_1)	110.00	Population Std. Deviation (σ_2)	125.00
5				
6	**Test Details**		**Excel Commands**	
7	Null Hypothesis (H_0) $\mu_1 - \mu_2 =$	0.00		
8	Alt. Hypothesis (H_1) $\mu_1 - \mu_2 \neq$	0.00		
9	Significance Level (α)	0.05		
10				
11	**Intermediate Calculations**			
12	Combined Variance	1109.0000	= ((B4^2) / B2) + ((D4^2) / D2)	
13	z test Statistic (z_{stat})	0.3094	= ((B3 - D3) - B7) / SQRT(B12)	
14				
15	**One-Tail Hypothesis Test**			
16	p-value	0.6963	= 2 * (1 - NORM.S.DIST(B13, TRUE))	
17				
18	**Decision**			
19	p-value approach	**Fail to reject H0**	= IF(B16<B9, "Reject H0", "Fail to reject H0")	

Die waarskynlikheid om steekproewe te kry met gemiddeldes wat met 13 uur $(375 - 362)$ of meer verskil wanneer die nulhipotese waar is, is 0.6963.
/ *The probability of obtaining samples whose means differ by* 13 *hours* $(375 - 362)$ *or more when the null hypothesis is true, is* 0.6963.

Probleem 3 / *Problem 3*

(a)

Nulhipotese / *Null Hypothesis* (H_0): $\mu_1 - \mu_2 = 0$
Alternatiewe Hipotese / *Alternative Hypothesis* (H_1): $\mu_1 - \mu_2 \neq 0$
Betekenisvlak / *Significance Level* (α): 0.05
Vryheidsgrade / *Degrees of Freedom* (df): 11+16-2=25
Hipotese Toets / *Hypothesis Test*: Tweestert / *Two-tail*
Kritieke waardes / *Critical Values*: $t_{crit} = \pm t_{0.025}^{(25)} = \pm 2.060$

Saamgevoegde-variansie / *Pooled variance*:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(11 - 1) \times 0.336^2 + (16 - 1) \times 0.140^2}{11 + 16 - 2} = 0.0569$$

Toetsstatistiek / *Test Statistic* (t_{stat}):

$$t_{stat} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(250.140 - 249.916) - 0}{\sqrt{0.0569\left(\frac{1}{11} + \frac{1}{16}\right)}} \approx 2.3975$$

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.
Kopiereg voorbehou / *Copyright reserved*

Beslissing: Aangesien $t_{stat} \approx 2.3975 > 2.060 = t_{krit}$, verwerp H_0. Daar is genoeg bewyse dat van 'n verskil in die populasiegemiddelde gewig van die vul van blikkies op die twee produksie lyne.

/ *Decision: Since* $t_{stat} \approx 2.3975 > 2.060 = t_{crit}$, *reject* H_0. *There is sufficient evidence of a difference in the population average weights of the cans filled on the two lines*.

(b)

	A	B	C	D
1	**Lyn A / *Line A***		**Lyn B / *Line B***	
2	Sample Size (n_1)	11	Sample Size (n_2)	16
3	Sample Mean $(\bar{x}_1)$	250.140	Sample Mean $(\bar{x}_2)$	249.916
4	Sample Standard Deviation (s_1)	0.336	Sample Standard Deviation (s_2)	0.140
5				
6	**Test Details**		**Excel Commands**	
7	Null Hypothesis (H_0) $\mu_1 - \mu_2 =$	0.00		
8	Alt. Hypothesis (H_1) $\mu_1 - \mu_2 \neq$	0.00		
9	Significance Level (α)	0.05		
10				
11	**Intermediate Calculations**			
12	Degrees of Freedom (df)	25	= B2 + D2 - 2	
13	Pooled Variance (s_p^2)	0.0569	= (((B2-1) * (B4^2)) + ((D2-1) * (D4^2))) / B12	
14	t test Statistic (t_{stat})	2.3972	= (B3 - D3) / SQRT(B13 * ((1/B2) + (1/D2)))	
15				
16	**One-Tail Hypothesis Test**			
17	p-value	0.0243	= 2 * (1 - T.DIST(ABS(B14), B12, TRUE))	
18				
19	**Decision**			
20	p-value approach	Reject H0	= IF(B17<B9, "Reject H0", "Fail to reject H0")	

Die waarskynlikheid om steekproewe te kry met gemiddeldes wat met $0.224g$ $(250.140 - 249.916)$ of meer verskil wanneer die nulhipotese waar is, is 0.0243.

/ *The probability of obtaining samples whose means differ by* $0.224g$ $(250.140 - 249.916)$ *or more when the null hypothesis is true, is* 0.0243.

Probleem 4 / *Problem 4*

$$\text{df} = n_d - 1 = 20 - 1 = 19$$

where, $n_d$is die aantal datapare. / *where.* n_d *is the number of pairs in data sample.*

Probleem 5 / *Problem 5*

$$\text{df} = n_d - 1 = 15 - 1 = 14$$

where, $n_d$is die aantal datapare. / *where.* n_d *is the number of pairs in data sample.*

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.

Kopiereg voorbehou / *Copyright reserved*

Probleem 6 / *Problem 6*

(a)

Handboek / *Textbook*	Boekwinkel / *Book Store*	*Amazon.com*	Difference
1	52.22	57.34	-5.12
2	52.74	44.47	8.27
3	39.04	41.48	-2.44
4	101.28	73.72	27.56
5	37.45	42.04	-4.59
6	113.41	95.38	18.03
7	109.72	119.80	-10.08
8	101.28	62.48	38.80
9	29.49	32.43	-2.94
10	70.07	74.43	-4.36
11	83.87	83.81	0.06
12	23.21	26.48	-3.27
13	72.80	73.48	-0.68
14	17.41	20.98	-3.57
15	37.72	40.43	-2.71
Mean ($\bar{x}_d$)			**3.5307**
Standard Deviation (s_d)			**13.8493**

Nulhipotese / *Null Hypothesis* | H_0: $\mu_d = 0$
Daar is geen verskil in die populasiegemiddelde prys van die handboeke tussen die plaaslike handelaar en Amazon.com nie.
/ *There is no difference in the population average price of textbooks between the local bookstore and Amazon.com.*

Alternatiewe Hipotese / *Alternative Hypothesis* | H_1: $\mu_d \neq 0$
Daar is 'n verskil in die populasiegemiddelde prys van die handboeke tussen die plaaslike handelaar en Amazon.com.
/ *There is a difference in the population average price of textbooks between the local bookstore and Amazon.com.*

Vryheidsgrade / *Degrees of Freedom* (df): $15 - 1 = 14$
Hipotese Toets / *Hypothesis Test*: Tweestert / *Two-tail*
Kritieke waardes / *Critical Values*: $t_{crit} = \pm t_{0.005}^{(14)} = \pm 2.977$

Toetsstatistiek / *Test Statistic* (t_{stat}):

$$t_{stat} = \frac{\bar{x}_d - \mu_d}{s_d/\sqrt{n_d}} = \frac{3.5307 - 0}{13.8493/\sqrt{15}} \approx 0.9874$$

Beslissing: Aangesien $t_{stat} \approx 0.9874 < 2.977 = t_{krit}$, verwerp nie H_0 nie. Daar is nie genoeg bewyse om tot die gevolgtrekking te kom dat daar 'n verskil is tussen die populasiegemiddeldes van die pryse van die handboeke by die plaaslike handelaar en Amazon.com is nie.

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.
Kopiereg voorbehou / *Copyright reserved*

/ <u>*Decision*</u>*: Since* $t_{stat} \approx 0.9874 < 2.977 = t_{crit}$*, do not reject* H_0*. There is not enough evidence to conclude that there is a difference in the population average price of textbooks between the local bookstore and Amazon.com*.

(b) Die aanname moet gemaak word dat die verdeling van die verskille tussen die pryse van die handboeke tussen die plaaslike handelaar en Amazon.com benaderd normaal is.
/ *One must assume that the distribution of the differences between the prices of textbooks between the two bookstores is approximately normal*.

(c) 99% vertrouensinterval / 99% *confidence interval*:

$$\bar{x}_d \pm t_{0.005}^{(14)} \times \frac{s_d}{\sqrt{n_d}} = 3.5307 \pm 2.977 \times \frac{13.8493}{\sqrt{15}}$$
$$= -7.1140 < \mu_d < 14.1754$$

Ons het 'n 99% vertroue dat die populasiegemiddelde verskil tussen die pryse, in die interval -7.1140 en 14.1754 lê.
/ *We are* 99% *confident that the population mean difference between the prices lies within the interval* -7.1140 *to* 14.1754.

(d) Die resultate in (a) en (c) is dieselfde. Die hipotetiese waarde van 0 vir die verskil tussen die populasiegemiddeldes is in die 99% vertrouensinterval ingesluit.
/ The results in (a) and (c) are the same. The hypothesised value of 0 *for the difference in the average population means is inside the* 99% *confidence interval.*

Probleem 7 / *Problem 7*

(a)

Nulhipotese / *Null Hypothesis* (H_0):	$\pi_1 - \pi_2 = 0$
Alternatiewe Hipotese / *Alternative Hypothesis* (H_1):	$\pi_1 - \pi_2 \neq 0$
Betekenisvlak / *Significance Level* (α):	0.05
Hipotese Toets / *Hypothesis Test*:	Tweestert / *Two-tail*
Kritieke waardes / *Critical Values*:	$z_{crit} = \pm z_{0.025} = \pm 1.96$

Steekproefverhoudings / *Sample proportions*:

$$p_1 = \frac{x_1}{n_1} = \frac{50}{100} = 0.50 \quad \text{en / } and \quad p_1 = \frac{x_2}{n_2} = \frac{30}{100} = 0.30$$

Saamgevoegde-proporsie / *Pooled proportion*:

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{50 + 30}{100 + 100} = 0.40$$

Toetsstatistiek / *Test Statistic*:

$$z_{stat} = \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(0.50 - 0.30) - 0}{\sqrt{0.40 \times 0.60 \times \left(\frac{1}{100} + \frac{1}{100}\right)}} \approx 2.8868$$

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.
Kopiereg voorbehou / *Copyright reserved*

Beslissing: Aangesien $z_{stat} = 2.8868 > 1.96 = z_{krit}$, verwerp H_0. Daar is genoeg getuienis om tot die gevolgtrekking te kom dat die populasie porporsie suksesse verskil vir Groep 1 en Groep 2.

/ *Decision: Since* $z_{stat} = 2.8868 > 1.96 = z_{crit}$, *reject* H_0. *There is sufficient evidence to conclude that there is a difference between the population proportion of successes for Group 1 and Group 2.*

	A	B	C	D
1	**Population 1**		**Population 2**	
2	Number of successes (x_1)	50	Number of successes (x_2)	30
3	Sample Size (n_1)	100	Sample Size (n_2)	100
4				
5	**Test Details**		**Excel Commands**	
6	Null Hypothesis (H_0) $\pi_1 - \pi_2 =$	0.00		
7	Alt. Hypothesis (H_1) $\pi_1 - \pi_2 \neq$	0.00		
8	Significance Level (α)	0.05		
9				
10	**Intermediate Calculations**			
11	Sample 1 Proportion (p_1)	0.5000	= B2 / B3	
12	Sample 2 Proportion (p_2)	0.3000	= D2 / D3	
13	Pooled Proportion $(\bar{p})$	0.4000	= (B2 + D2) / (B3 + D3)	
14	Standard Error	0.0693	= SQRT(B13 * (1-B13) * ((1/B3) + (1/D3)))	
15	z test Statistic (z_{stat})	2.8868	= ((B11 - B12) - B6) / B14	
16				
17	**Two-Tail Hypothesis Test**			
18	p-value	0.0039	= 2 * (1 - NORM.S.DIST(ABS(B15), TRUE))	
19				
20	**Decision**			
21	p-value approach	Reject H0	= IF(B18<B8, "Reject H0", "Fail to reject H0")	

Beslissing: Aangesien $p\text{-}waarde = 0.0039 < 0.05 = \alpha$, verwerp H_0. Daar is genoeg getuienis om tot die gevolgtrekking te kom dat die populasie porporsie suksesse verskil vir Groep 1 en Groep 2.

/ *Decision: Since* $p\text{-}value = 0.0039 < 0.05 = \alpha$, *reject* H_0. *There is sufficient evidence to conclude that there is a difference between the population proportion of successes for Group 1 and Group 2.*

(b) 95% vertrouensinterval / 95% *confidence interval*:

$$(p_1 - p_2) \pm z_{0.025} \times \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} =$$

$$(0.50 - 0.30) \pm 1.96 \times \sqrt{\frac{0.50(1-0.50)}{100} + \frac{0.30(1-0.30)}{100}} =$$

$$0.0671 < \pi_1 - \pi_2 < 0.3329$$

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.

Kopiereg voorbehou / *Copyright reserved*

Probleem 8 / *Problem 8*

(a)

Nulhipotese / *Null Hypothesis* (H_0): $\pi_1 - \pi_2 = 0$
Alternatiewe Hipotese / *Alternative Hypothesis* (H_1): $\pi_1 - \pi_2 \neq 0$
Betekenisvlak / *Significance Level* (α): 0.01
Hipotese Toets / *Hypothesis Test*: Tweestert / *Two-tail*
Kritieke waardes / *Critical Values*: $z_{crit} = \pm z_{0.005} = \pm 2.575$

Steekproefverhoudings / *Sample proportions*:

$$p_1 = \frac{x_1}{n_1} = \frac{45}{100} = 0.45 \quad \text{en / } and \quad p_1 = \frac{x_2}{n_2} = \frac{25}{50} = 0.50$$

Saamgevoegde-proporsie / *Pooled proportion*:

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{45 + 25}{100 + 50} = 0.4667$$

Toetsstatistiek / *Test Statistic*:

$$z_{stat} = \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(0.45 - 0.50) - 0}{\sqrt{0.4667 \times 0.5333 \times \left(\frac{1}{100} + \frac{1}{50}\right)}} \approx -0.5786$$

Beslissing: Aangesien $z_{stat} = -0.5786 > -1.96 = z_{krit}$, verwerp nie H_0. Daar is onvoldoende bewyse dat die populasie proporsies vir sukses tussen groepe 1 en 2 verskil.

/ *Decision: Since* $z_{stat} = -0.5786 > -1.96 = z_{crit}$, *fail to reject* H_0. *There is insufficient evidence to conclude that there is a difference between the population proportion of successes for Group 1 and Group 2.*

	A	B	C	D
1	**Population 1**		**Population 2**	
2	Number of successes (x_1)	45	Number of successes (x_2)	25
3	Sample Size (n_1)	100	Sample Size (n_2)	50
4				
5	**Test Details**		**Excel Commands**	
6	Null Hypothesis (H_0) $\pi_1 - \pi_2 =$	0.00		
7	Alt. Hypothesis (H_1) $\pi_1 - \pi_2 \neq$	0.00		
8	Significance Level (α)	0.01		
9				
10	**Intermediate Calculations**			
11	Sample 1 Proportion (p_1)	0.4500	= B2 / B3	
12	Sample 2 Proportion (p_2)	0.5000	= D2 / D3	
13	Pooled Proportion ($\bar{p}$)	0.4667	= (B2 + D2) / (B3 + D3)	
14	Standard Error	0.0864	= SQRT(B13 * (1-B13) * ((1/B3) + (1/D3)))	
15	z test Statistic (z_{stat})	-0.5786	= ((B11 - B12) - B6) / B14	
16				
17	**Two-Tail Hypothesis Test**			
18	p-value	0.5628	= 2 * (1 - NORM.S.DIST(ABS(B15), TRUE))	
19				
20	**Decision**			
21	p-value approach	Fail to reject H0	= IF(B18<B8, "Reject H0", "Fail to reject H0")	

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.
Kopiereg voorbehou / *Copyright reserved*

Beslissing: Aangesien $p\text{-}waarde = 0.5628 > 0.01 = \alpha$, verwerp nie H_0. Daar is onvoldoende bewyse dat die populasie proporsies vir sukses tussen groepe 1 en 2 verskil.

/ *Decision: Since* $p\text{-}value = 0.5628 > 0.01 = \alpha$, *fail to reject* H_0. *There is insufficient evidence to conclude that there is a difference between the population proportion of successes for Group 1 and Group 2.*

(b) 99% vertrouensinterval / 99% *confidence interval*:

$$(p_1 - p_2) \pm z_{0.005} \times \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} =$$

$$(0.45 - 0.50) \pm 2.575 \times \sqrt{\frac{0.45(1-0.45)}{100} + \frac{0.50(1-0.50)}{50}} =$$

$$-0.2726 < \pi_1 - \pi_2 < 0.1726$$

Probleem 9 / *Problem 9*

Laat Populasie 1 = Mans & Populasie 2 = Dames.
/ *Let Population 1= Males & Population 2 = Females.*

(a)

Nulhipotese / *Null Hypothesis* (H_0):	$\pi_1 - \pi_2 = 0$
Alternatiewe Hipotese / *Alternative Hypothesis* (H_1):	$\pi_1 - \pi_2 < 0$
Betekenisvlak / *Significance Level* (α):	0.01
Hipotese Toets / *Hypothesis Test*:	Eenstert / *One-tail*
Kritieke waarde / *Critical Value*:	$z_{crit} = z_{0.01} = -2.33$

Steekproefverhoudings / *Sample proportions*:

$$p_1 = \frac{x_1}{n_1} = \frac{136}{240} = 0.5667 \quad \text{en / } and \quad p_1 = \frac{x_2}{n_2} = \frac{224}{260} = 0.8615$$

Saamgevoegde-proporsie / *Pooled proportion*:

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{136 + 224}{240 + 260} = 0.7200$$

Toetsstatistiek / *Test Statistic*:

$$z_{stat} = \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(0.5667 - 0.8615) - 0}{\sqrt{0.72 \times (1 - 0.72) \times \left(\frac{1}{240} + \frac{1}{260}\right)}} \approx -7.3348$$

Beslissing: Aangesien $z_{stat} = -7.3348 < -2.33 = z_{krit}$, word H_0. Daar is genoeg bewyse om tot die gevolgtrekking te kom dat die populasie proporsies mans wat daarvan hou om te klere te koop beduidend kleiner is as dié van dames.

/ *Decision: Since* $z_{stat} = -7.3348 < -2.33 = z_{crit}$, *reject* H_0. *There is sufficient evidence to conclude that the population proportion of males who enjoy shopping for clothing is significantly smaller than that of females.*

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.
Kopiereg voorbehou / *Copyright reserved*

(b)

	A	B	C	D
1	**Male Population**		**Female Population**	
2	Enjoy cloth shopping (x_1)	136	Enjoy cloth shopping (x_2)	224
3	Sample Size (n_1)	240	Sample Size (n_2)	260
4				
5	**Test Details**		**Excel Commands**	
6	Null Hypothesis (H_0) $\pi_1 - \pi_2 =$	0.00		
7	Alt. Hypothesis (H_1) $\pi_1 - \pi_2 <$	0.00		
8	Significance Level (α)	0.01		
9				
10	**Intermediate Calculations**			
11	Sample 1 Proportion (p_1)	0.5667	= B2 / B3	
12	Sample 2 Proportion (p_2)	0.8615	= D2 / D3	
13	Pooled Proportion ($\bar{p}$)	0.7200	= (B2 + D2) / (B3 + D3)	
14	Standard Error	0.0402	= SQRT(B13 * (1-B13) * ((1/B3) + (1/D3)))	
15	z test Statistic (z_{stat})	-7.3366	= ((B11 - B12) - B6) / B14	
16				
17	**One-Tail Hypothesis Test**			
18	p-value	0.0000	= 1 - NORM.S.DIST(ABS(B15), TRUE)	
19				
20	**Decision**			
21	p-value approach	Reject H0	= IF(B18<B8, "Reject H0", "Fail to reject H0")	

Beslissing: Aangesien $p\text{-}waarde = 0.0000 < 0.01 = \alpha$, word H_0. Die waarskynlikheid dat die steekproef proporsie dames meer as 0.2948 $(0.5667 - 0.8615)$ groter as dié van mans is, indien die nulhipotese waar is, is so te sê nul. Daar is onvoldoende bewyse dat die populasie proporsies vir sukses tussen groepe 1 en 2 verskil.

/ *Decision: Since* $p\text{-}value = 0.0000 < 0.01 = \alpha$, *reject* H_0. *The probability of obtaining a sample proportion of females more than* 0.2948 $(0.5667 - 0.8615)$ *larger than that of males when the null hypothesis is true, is virtually zero. Therefore, there is sufficient evidence to conclude that the population proportion of males who enjoy shopping for clothing is significantly smaller than that of females*.

(c) 99% vertrouensinterval / 99% *confidence interval*:

$$(p_1 - p_2) \pm z_{0.005} \times \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} =$$

$$(0.5667 - 0.8615) \pm 2.575 \times \sqrt{\frac{0.5667(1-0.5667)}{240} + \frac{0.8615(1-0.8615)}{260}} =$$

$$-0.3939 < \pi_1 - \pi_2 < -0.1957$$

(d) Die resultate kan nie vergelyk word nie, aangesien die hipotese toets in (a) eenkantig is.

/ *The results cannot be compared since the hypothesis test in (a) is one-sided*.

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.

Kopiereg voorbehou / *Copyright reserved*

Probleem 10 / *Problem 10*

Laat Populasie 1 = 2 tot 7 jaar & Populasie 2 = 8 tot 18 jaar.
/ *Let Population 1= 2 to 7 years & Population 2 = 8 to 18 years of age.*

(a)

Nulhipotese / *Null Hypothesis* (H_0):	$\pi_1 - \pi_2 = 0$
Alternatiewe Hipotese / *Alternative Hypothesis* (H_1):	$\pi_1 - \pi_2 \neq 0$
Betekenisvlak / *Significance Level* (α):	0.05
Hipotese Toets / *Hypothesis Test*:	Tweetstert / *Two-tail*
Kritieke waardes / *Critical Values*:	$z_{crit} = \pm z_{0.025} = \pm 1.96$

Steekproefverhoudings / *Sample proportions*:

$$p_1 = \frac{x_1}{n_1} = \frac{283}{1090} = 0.2596 \quad \text{en / } and \quad p_1 = \frac{x_2}{n_2} = \frac{1053}{2065} = 0.5099$$

Saamgevoegde-proporsie / *Pooled proportion*:

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{283 + 1053}{1090 + 2065} = 0.4235$$

Toetsstatistiek / *Test Statistic*:

$$z_{stat} = \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(0.2596 - 0.5099) - 0}{\sqrt{0.4235 \times (1 - 0.4235) \times \left(\frac{1}{1090} + \frac{1}{2065}\right)}} \approx -13.5303$$

<u>Beslissing</u>: Aangesien $z_{stat} = -13.5303 < -1.96 = z_{krit}$, verwerp H_0. Daar is genoeg bewyse om tot die gevolgtrekking te kom dat daar 'n betekenisvolle verskil is tussen die proporsie kinders in die ouderdomsgroep 2 tot 7 jaar en die groep 8 tot 18 jaar wat daagliks 'n rekenaar gebruik.

/ <u>*Decision*</u>*: Since* $z_{stat} = -13.5303 < -1.96 = z_{crit}$*, reject* H_0*. There is sufficient evidence to conclude that a significant difference exists in the proportion of children between the ages of 2 and 7 years, and between the age of 8 and 18 years who use a computer each day*.

(b)

<u>Beslissing</u>: Aangesien $p\text{-}waarde = 0.0000 < 0.05 = \alpha$, word H_0. Die waarskynlikheid om 'n verskil so groot soos 0.2503 (0.2596 − 0.5099) of meer tussen twee steekproef proporsies te kry indien die nulhipotese waar is, is so te sê nul. Daar is genoeg bewyse om tot die gevolgtrekking te kom dat daar 'n betekenisvolle verskil is tussen die proporsie kinders in die ouderdomsgroep 2 tot 7 jaar en die groep 8 tot 18 jaar wat daagliks 'n rekenaar gebruik.

/ <u>*Decision*</u>*: Since* $p\text{-}value = 0.0000 < 0.05 = \alpha$*, reject* H_0*. The probability of obtaining a difference in the two sample proportions as large as* 0.2503 *(*0.2596 – 0.5099*) or more is virtually* 0 *if the null hypothesis is true. Therefore, there is sufficient evidence to conclude that a significant difference exists in the proportion of children between the ages of 2 and 7 years, and between the age of 8 and 18 years who use a computer each day*.

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.
Kopiereg voorbehou / *Copyright reserved*

	A	B	C	D
1	**Population of 2-7-year-olds**		**Population of 8-18-year-olds**	
2	Use Computer Daily (x_1)	283	Use Computer Daily (x_2)	1053
3	Sample Size (n_1)	1090	Sample Size (n_2)	2065
4				
5	**Test Details**		**Excel Commands**	
6	Null Hypothesis (H_0) $\pi_1 - \pi_2 =$	0.00		
7	Alt. Hypothesis (H_1) $\pi_1 - \pi_2 \neq$	0.00		
8	Significance Level (α)	0.05		
9				
10	**Intermediate Calculations**			
11	Sample 1 Proportion (p_1)	0.2596	= B2 / B3	
12	Sample 2 Proportion (p_2)	0.5099	= D2 / D3	
13	Pooled Proportion ($\bar{p}$)	0.4235	= (B2 + D2) / (B3 + D3)	
14	Standard Error	0.0185	= SQRT(B13 * (1-B13) * ((1/B3) + (1/D3)))	
15	z test Statistic (z_{stat})	-13.5302	= ((B11 - B12) - B6) / B14	
16				
17	**Two-Tail Hypothesis Test**			
18	p-value	0.0000	= 2 * (1 - NORM.S.DIST(ABS(B15), TRUE))	
19				
20	**Decision**			
21	p-value approach	Reject H0	= IF(B18<B8, "Reject H0", "Fail to reject H0")	

(c) 95% vertrouensinterval / 95% *confidence interval*:

$$(p_1 - p_2) \pm z_{0.025} \times \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} =$$

$$(0.2596 - 0.5099) \pm 1.96 \times \sqrt{\frac{0.2596(1-0.2596)}{1090} + \frac{0.5099(1-0.5099)}{2065}} =$$

$$-0.2841 < \pi_1 - \pi_2 < -0.2165$$

Probleem 11 / *Problem 11*

(a) $F_{0.05}^{(16-1,21-1)} = F_{0.05}^{(15,20)} = 2.20$ (b) $F_{0.01}^{(16-1,21-1)} = F_{0.01}^{(15,20)} = 3.09$

(c) $F_{0.05}^{(10-1,10-1)} = F_{0.05}^{(9,9)} = 3.18$ (d) $F_{0.01}^{(10-1,10-1)} = F_{0.01}^{(9,9)} = 5.35$

Probleem 12 / *Problem 12*

(a) $F_{0.10/2}^{(16-1,21-1)} = F_{0.05}^{(15,20)} = 2.20$ (b) $F_{0.05/2}^{(16-1,21-1)} = F_{0.025}^{(15,20)} = 2.57$

(c) $F_{0.02/2}^{(16-1,21-1)} = F_{0.01}^{(15,20)} = 3.09$ (d) $F_{0.10/2}^{(10-1,10-1)} = F_{0.05}^{(9,9)} = 3.18$

(e) $F_{0.05/2}^{(10-1,10-1)} = F_{0.025}^{(9,9)} = 4.03$ (f) $F_{0.02/2}^{(10-1,10-1)} = F_{0.01}^{(9,9)} = 5.35$

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.
Kopiereg voorbehou / *Copyright reserved*

Probleem 13 / *Problem 13*

Laat Populasie 1 = "Nie-hoofvak" populasie & Populasie 2 = "Hoofvak" populasie.
/ *Let Population 1= "Non-major" population & Population 2 = "Major" population.*

(a)

Nulhipotese / *Null Hypothesis* (H_0):	$\sigma_1^2 = \sigma_2^2$
Alternatiewe Hipotese / *Alternative Hypothesis* (H_1):	$\sigma_1^2 > \sigma_2^2$
Betekenisvlak / *Significance Level* (α):	0.05
Hipotese Toets / *Hypothesis Test*:	Eenstert / *One-tail*
Kritieke waarde / *Critical Value*:	$F_{crit} = F_{0.05}^{(12,9)} = 3.07$

Toetsstatistiek / *Test Statistic*:

$$F_{stat} = \frac{s_1^2}{s_2^2} = \frac{210.2}{36.5} \approx 5.7589$$

Beslissing: Aangesien $F_{stat} = 5.7589 > 3.07 = F_{krit}$, verwerp H_0. Daar is genoeg bewyse dat die populasie variansie in die finale eksamenpunte van die "Nie-hoofvak" populasie meer varieer as die eksamenpunte van dié van die "Hoofvak" populasie.

/ *Decision: Since $F_{stat} = 5.7589 > 3.07 = F_{crit}$, reject H_0. There is sufficient evidence to conclude that the population variance in final exam scores of "Non-major" students is greater than the population variance in final exam scores of accounting "Major" students.*

(b) Die toets vereis dat die twee populasies normaal verdeel is, en dat die steekproewe onafhanklik en ewekansig geneem is.
/ *The test assumes that the two populations are each normally distributed, and that the samples are independent and random in each case.*

Probleem 14 / *Problem 14*

(a)

Nulhipotese / *Null Hypothesis* (H_0):	$\sigma_1^2 = \sigma_2^2$
Alternatiewe Hipotese / *Alternative Hypothesis* (H_1):	$\sigma_1^2 > \sigma_2^2$
Betekenisvlak / *Significance Level* (α):	0.05
Hipotese Toets / *Hypothesis Test*:	Eenstert / *One-tail*
Kritieke waarde / *Critical Value*:	$F_{crit} = F_{0.05}^{(14,19)} = 2.26$

Toetsstatistiek / *Test Statistic*:

$$F_{stat} = \frac{s_1^2}{s_2^2} = \frac{18.9^2}{15^2} \approx 1.5876$$

Beslissing: Aangesien $F_{stat} = 1.5876 < 2.26 = F_{krit}$, verwerp nie H_0. Daar is onvoldoende bewyse dat die populasie variansie van die Century Equity Growth fonds groter is as dié van die Fidelity Growth fonds.

/ *Decision: Since $F_{stat} = 1.5876 < 2.26 = F_{crit}$, fail to reject H_0. There is insufficient evidence to conclude that the population variance of the Century Equity Growth fund is larger than that of the Fidelity Growth fund.*

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.
Kopiereg voorbehou / *Copyright reserved*